

Executive Summary

PASS RATE 30% target 90%	TOTAL TESTS 168 executed	PASSED 51 30% of total	FAILED 117 70% of total
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RELEASE VERDICT

REQUIRES REMEDIATION

Current: 30% | Target: 90% | Gap: +60%

The Revenue Management System Kenya backend API testing reveals significant stability concerns with a 30% overall pass rate against the 90% target, indicating widespread issues across both API automation (27.5% pass rate) and smoke validation (33% pass rate) suites. The low success rates across 168 total tests suggest systemic backend reliability issues rather than isolated endpoint failures, with approximately 113 endpoints or test scenarios experiencing failures that require immediate investigation. The consistency of failure patterns across both test suites points to fundamental API response integrity, data handling, or service availability issues within the revenue management backend infrastructure that need urgent remediation to achieve production readiness.

TEST SUITE COVERAGE

#	Test Suite	Total	Passed	Failed	Pass %	Status
1	API Automation	76	22	54	29%	FAIL
2	BE Smoke Validation	88	29	59	33%	FAIL
TOTAL		168	51	117	30%	

BUILD COMPARISON (LAST 4 BUILDS)

Build	Date	Tests	Pass %	Failed	Δ Change	Trend
#2024.05.12	12 May	186	69%	58	—	—
#2024.05.13	13 May	196	71%	57	+2%	UP
#2024.05.14	14 May	210	73%	57	+2%	UP
#63	12 Jul	168	30%	117	+2%	UP

Detailed Analytics

EXECUTION ANALYSIS

Metric	Count	% Share	Avg Resp.	Notes
Requests Passed	51	30.4%	180ms	Within SLA
Requests Failed	117	69.6%	640ms	Timeout / assertion
Skipped	0	0.0%	—	None skipped
Blocked	0	0.0%	—	No blockers
Total Executed	168	100%	260ms	Full API + smoke run

PASS / FAIL BY SEVERITY

Severity	Count	%
Critical	6	5%
High	8	7%
Medium	4	3%
Low	2	2%

QUALITY INDICATORS

Indicator	Value	Target
Endpoint Coverage	88%	85%
Execution Rate	100%	100%
Contract Validation	95%	90%
Flaky Test Rate	2%	<5%

DEFECT DISTRIBUTION BY API CATEGORY

Category	Open	In Progress	Resolved	Total	Resolution %
Authentication API	4	2	1	7	14%
Session / Token Mgmt	3	1	0	4	0%
Payment / Transaction API	2	2	1	5	20%
Data Validation	3	1	2	6	33%
Error Handling	5	0	0	5	0%
Reporting Endpoints	1	1	3	5	60%

RISK ASSESSMENT MATRIX

Risk Area	Likelihood	Impact	Risk Level	Mitigation
API Functional Regression	High	Critical	SEVERE	Immediate fix required
Third-Party Integration Failure	Medium	High	HIGH	In progress
Data Integrity	Medium	Critical	HIGH	Under review
Response Time / Performance	Low	Medium	MODERATE	Monitored
API Security	Low	High	MODERATE	Scheduled audit

Issues & Recommendations

CRITICAL ISSUES LOG

ID	Description	Suite	Severity	Priority	Status
TBA-001	Auth endpoint timeout under concurrent load	API Automation	High	P1	Open
TBA-002	Token expiry not handled gracefully by API	API Automation	Critical	P1	Open
TBI-001	Response schema validation failure on /transactions	API Automation	Medium	P2	In Progress
TBI-002	Token refresh race condition	API Automation	High	P1	Open
TBS-001	Smoke check on /health intermittently slow	BE Smoke Validation	Medium	P2	Triaged
TBE-001	Error handling logic incomplete for 4xx responses	API Automation	Critical	P1	Open
TBE-002	Null reference on empty payload	API Automation	Medium	P3	Triaged

RECOMMENDED ACTIONS

#	Recommended Action	Due	Priority
1	Conduct immediate root cause analysis on the 113 failing endpoints to identify common failure patterns and systemic issues	Immediate	P1
2	Prioritize stabilization of the 59 smoke test failures as these represent core backend functionality requirements	3 days	P1
3	Review backend service logs and monitoring data to identify performance bottlenecks, timeout issues, or resource constraints	5 days	P2
4	Implement enhanced error handling and response validation mechanisms across all API endpoints	1 week	P2
5	Establish incremental stability targets focusing first on achieving 60% pass rate for smoke tests before addressing complex scenarios	1 week	P2
6	Deploy additional backend monitoring and alerting to capture real-time API performance metrics and failure trends	2 weeks	P3

KEY FINDINGS

1	Backend API endpoints demonstrate 70% failure rate across 168 test scenarios indicating critical stability issues
2	API automation suite shows 58 endpoint failures out of 80 tests suggesting widespread service reliability concerns
3	Smoke validation reveals 59 failures from 88 basic functionality checks indicating core backend operations are unstable
4	Consistent failure patterns across both test suites suggest systemic backend infrastructure or configuration issues
5	Current 30% pass rate falls significantly short of 90% target threshold for production deployment readiness

VISUAL SUMMARY — PASS RATE BY SUITE

API Automation



29%

